# Linux Health Monitoring Tool

A Project-Based Learning Initiative | Information Technology, Semester 6


## By:

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## Branch & Semester

Information Technology, Sem-6

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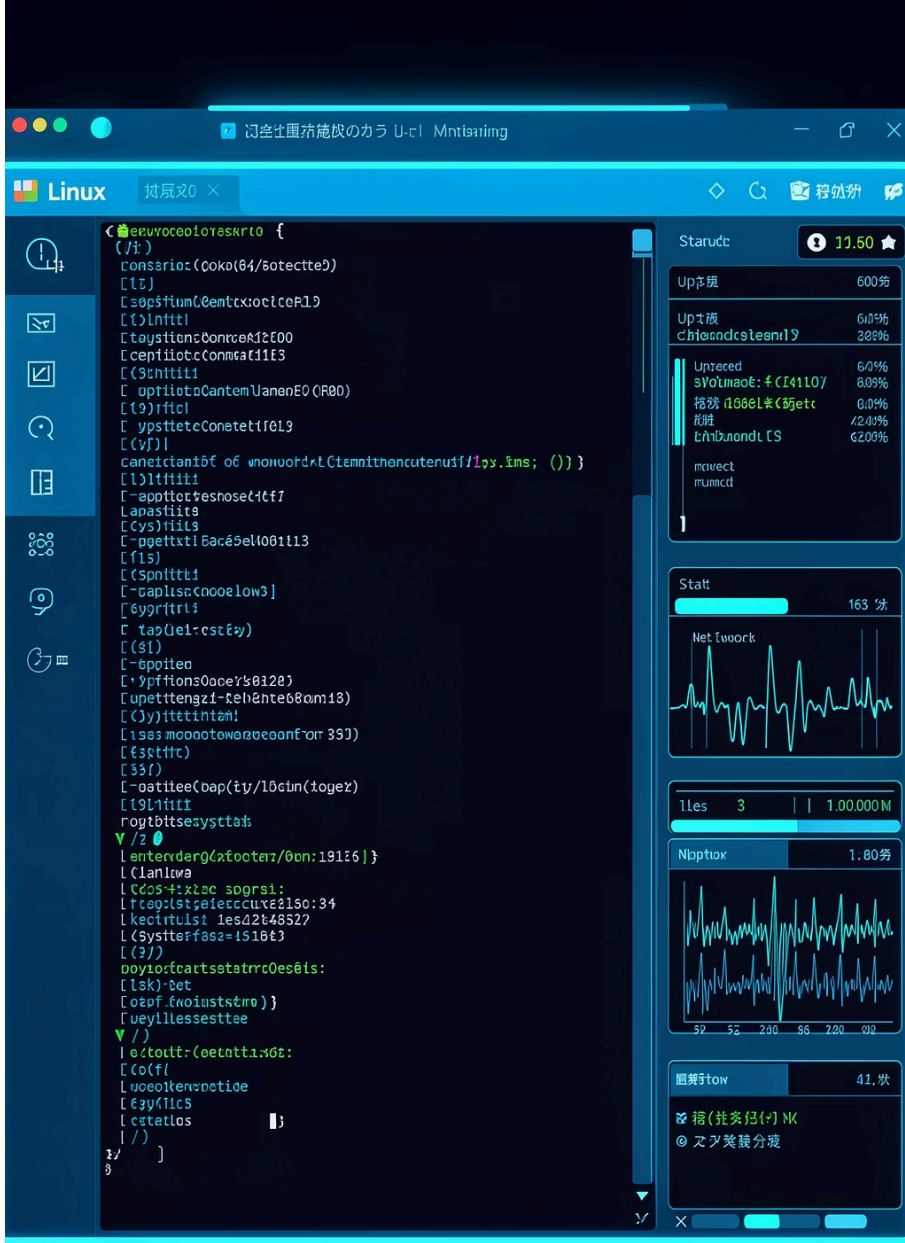


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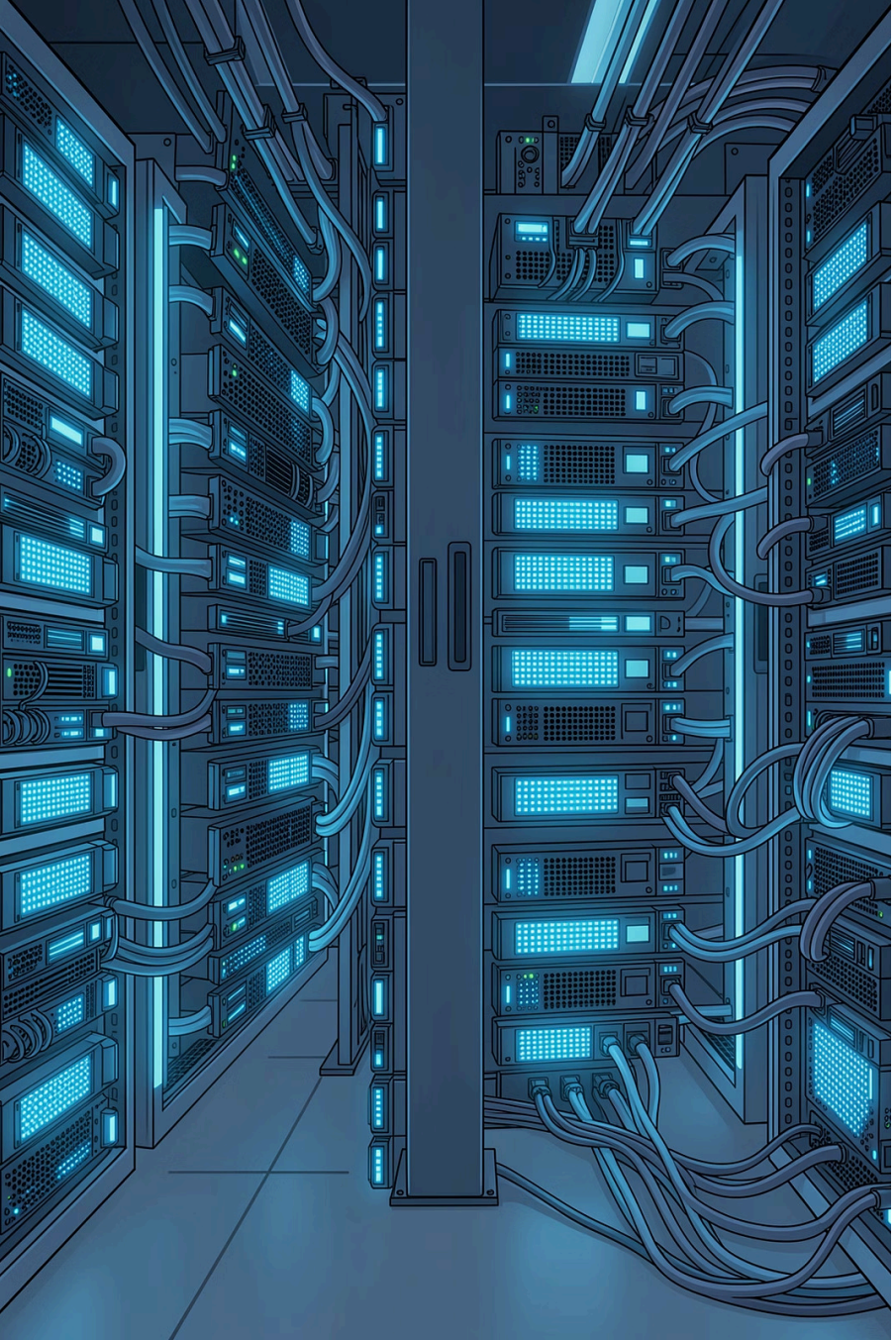

# Introduction

In modern computing environments, **system health monitoring** is critical to ensuring stability, performance, and availability of Linux-based servers and workstations.

## The Challenge

Manual monitoring using individual commands is time-consuming, error-prone, and unsustainable at scale. Administrators often miss critical performance thresholds.

## The Solution

Automating system monitoring through **Bash shell scripting** enables continuous, consistent, and structured data collection — without manual intervention.

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# Project Objectives

### Monitor System Resources

Track CPU usage, memory consumption, and disk utilization in real time using native Linux utilities.

### Automate Logging

Automate the collection of performance data into structured, timestamped log files.

### Modular Architecture

Different modules for different functions.

### Learn System-Level Linux

Develop hands-on proficiency with Linux internals, shell scripting, and system administration concepts.

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# Tools & Technologies Used

### Linux OS

Ubuntu / Debian-based environment

### Bash Shell Scripting

Core automation language for the tool

### Key Commands

top, free, df, awk, grep

### Git & GitHub

Version control and project hosting

### VS Code / Terminal Editors

Vim, Nano, and Visual Studio Code

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# System Design & Architecture

The tool follows a linear pipeline: the user initiates the Bash script, which sequentially collects resource metrics, processes raw command output into readable values, appends them to a timestamped log, and presents a clean summary in the terminal.

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# Working & Methodology

## Data Collection

The script invokes `top -bn1` for CPU, `free -m` for memory, and `df -h` for disk — each returning raw system metrics at the moment of execution.

## Processing & Parsing

Raw outputs are piped through `awk` and `grep` to extract specific fields — percentage usage, available space, and load averages.

## Logging

Processed values are appended to a `.log` file with a `date`-generated timestamp, building a chronological performance history.

## Output Display

A formatted summary is printed to the terminal, giving the user an at-a-glance view of current system health status.

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# Key Code Snippets

The following excerpts highlight the core logic behind each monitoring module. Full scripts are available on the GitHub repository.

### CPU Usage

```
#!/bin/bash

echo "
 CPU Load:"
top -bn1 | grep "load average"

source utils/formatter.sh

echo " CPU Load:"
load=$(top -bn1 | grep "load average" | awk -F'load
average:' '{print $2}')

print_ok "Load Average:$load"PU=$(top -bn1 | grep
"Cpu(s)" \
  | awk '{print $2}')
echo "CPU Usage: $CPU%"
```

### Memory Usage

```
#!/bin/bash

echo "
 Memory Usage:"
free -h
```

### Disk Usage

```
#!/bin/bash

echo "
Disk Usage:"
df -h
```

### Processes

```
#!/bin/bash

echo "
Top 5 CPU Processes:"
ps -eo pid,ppid,cmd,%cpu,%mem --sort=-%cpu | head -n 6

echo ""
echo " Top 5 Memory Processes:"
ps -eo pid,ppid,cmd,%cpu,%mem --sort=-%mem | head -n
6
```

### Monitoring.sh

```
#!/bin/bash

# Load formatter (for colors & structured output)
source utils/formatter.sh

# Log file
LOG_FILE="logs/system.log"

# Ensure logs directory exists
mkdir -p logs

# Header
print_header " Linux Health Monitor"

"
echo -e "\n

 Date: $(date)"
echo " Hostname: $(hostname)"

echo -e "\n Uptime:"
uptime

echo ""

# ========================
# CLI Argument Handling
# ========================

case $1 in

  --cpu)
    bash modules/cpu.sh
    ;;

  --memory)
    bash modules/memory.sh
    ;;

  --disk)
    bash modules/disk.sh
    ;;

  --process)
    bash modules/processes.sh
    ;;

  --alerts)
    bash modules/alerts.sh
    ;;

  --all | "")
    bash modules/cpu.sh
    echo ""
    bash modules/memory.sh
    echo ""
    bash modules/disk.sh
    echo ""
    bash modules/processes.sh
    echo ""
    bash modules/alerts.sh
    ;;

  --help)
    echo "Usage: ./monitor.sh [OPTION]"
    echo "--cpu       Show CPU usage"
    echo "--memory    Show memory usage"
    echo "--disk      Show disk usage"
    echo "--process   Show processes"
    echo "--alerts    Show alerts"
    echo "--all       Run full monitor"
    echo "--help      Show this help message"
    exit 0
    ;;

  *)
    echo " Invalid option"
    echo "Use --help to see available options"
    exit 1
    ;;

esac

# ========================
# Logging Section (FIXED)
# ========================

{
echo "==============================="
echo "Run at: $(date)"
echo "==============================="

uptime
free -h
df -h

echo ""

} >> "$LOG_FILE
```

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# Output & Results

## Terminal Output

```
📅 Date: Thu Apr 23 11:29:59 AM IST 2026
🖥️Hostname: nira-v-sasustuflinux

⏳ Uptime:
 11:29:59 up 2 min,  1 user,  load average: 0.23, 0.18, 0.08

⚙️CPU Load:
top - 11:29:59 up 2 min,  1 user,  load average: 0.23, 0.18, 0.08
⚙️CPU Load:
[OK] Load Average: 0.23, 0.18, 0.08

💾 Memory Usage:
              total        used        free      shared  buff/cache   available
Mem:           7.5Gi       1.5Gi       5.1Gi       279Mi       1.3Gi       5.9Gi
Swap:          4.0Gi          0B       4.0Gi

📀 Disk Usage:
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           767M  2.3M  764M   1% /run
/dev/nvme0n1p7   98G   20G   74G  21% /
tmpfs           3.8G     0  3.8G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
efivarfs        192K  136K   52K  73% /sys/firmware/efi/efivars
/dev/nvme0n1p1  256M   55M  202M  22% /boot/efi
tmpfs           767M  124K  766M   1% /run/user/1000

🔥 Top 5 CPU Processes:
    PID    PPID CMD                         %CPU %MEM
   2444    2154 /usr/bin/gnome-shell         4.5  4.4
   3570    2154 /usr/libexec/gnome-terminal  1.5  0.7
   2773    2598 /usr/libexec/ibus-extension  1.3  0.3
      1       0 /sbin/init splash            1.0  0.1
   3292    2154 /usr/bin/nautilus --gapplic  0.9  2.4

🧠 Top 5 Memory Processes:
    PID    PPID CMD                         %CPU %MEM
   2444    2154 /usr/bin/gnome-shell         4.5  4.4
   3292    2154 /usr/bin/nautilus --gapplic  0.9  2.4
   3034    2444 gjs /usr/share/gnome-shell/  0.4  0.9
   3344    2154 /usr/bin/gnome-calendar --g  0.3  0.8
   3570    2154 /usr/libexec/gnome-terminal  1.5  0.7

🚨 Disk Usage Alerts:
```

## Log File Entries

```
==============================
Run at: Thu Apr 23 11:29:58 AM IST 2026
==============================
 11:29:58 up 1 min,  1 user,  load average: 0.25, 0.19, 0.08
              total        used        free      shared  buff/cache   available
Mem:           7.5Gi       1.5Gi       5.1Gi       279Mi       1.3Gi       5.9Gi
Swap:          4.0Gi          0B       4.0Gi
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           767M  2.3M  764M   1% /run
/dev/nvme0n1p7   98G   20G   74G  21% /
tmpfs           3.8G     0  3.8G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
efivarfs        192K  136K   52K  73% /sys/firmware/efi/efivars
/dev/nvme0n1p1  256M   55M  202M  22% /boot/efi
tmpfs           767M  124K  766M   1% /run/user/1000

==============================
Run at: Thu Apr 23 11:29:59 AM IST 2026
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Filesystem      Size  Used Avail Use% Mounted on
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/dev/nvme0n1p7   98G   20G   74G  21% /
tmpfs           3.8G     0  3.8G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
efivarfs        192K  136K   52K  73% /sys/firmware/efi/efivars
/dev/nvme0n1p1  256M   55M  202M  22% /boot/efi
tmpfs           767M  124K  766M   1% /run/user/1000

nirav-panchal@nira-v-sasustuflinux:~/linux-health-monitor$
```

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# Conclusion

## Project Achievements

- This project successfully automated Linux system health monitoring using pure Bash scripting, creating a lightweight solution for tracking essential system performance metrics.
- It strengthened practical skills in Linux system administration, shell scripting, command-line tooling, and structured log generation for ongoing analysis and troubleshooting.
- The approach has clear real-world value in server monitoring, preventive maintenance, and early detection of performance issues that can affect reliability and uptime.
- Future enhancements could include alert notifications, support for additional metrics such as network and process usage, log rotation, and more advanced reporting or visualization.

Overall, system monitoring is essential for maintaining stable and secure computing environments, because timely visibility into system behavior enables informed decisions, faster responses, and better long-term performance.

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